The Mind Behind Linux | Linus Torvalds - TED.

Introduction Lecture

DV1457 / DV1578 Programming in UNIX Environments

Diego Navarro, Tek.Lic.
BTH

2024

Program

- Course Staff.
- Basic Information of the Course.
- Pedagogical Approach.
- Closing remarks.


Course Staff

Hi! I'm Diego!

- From Bogota, Colombia.
- Adjunct lecturer / Ph.D. Candidate.

- B.Eng. Multimedia Engineering (2009).
- M.Sc. Multimedia Management (2011).
- M.Sc. Design, Interaction and Game Technologies (2014).
- Tek.Lic. Computer Science (2020).
- Ph.D. Candidate in Computer Science (2021-2024).

Hi! I'm Diego

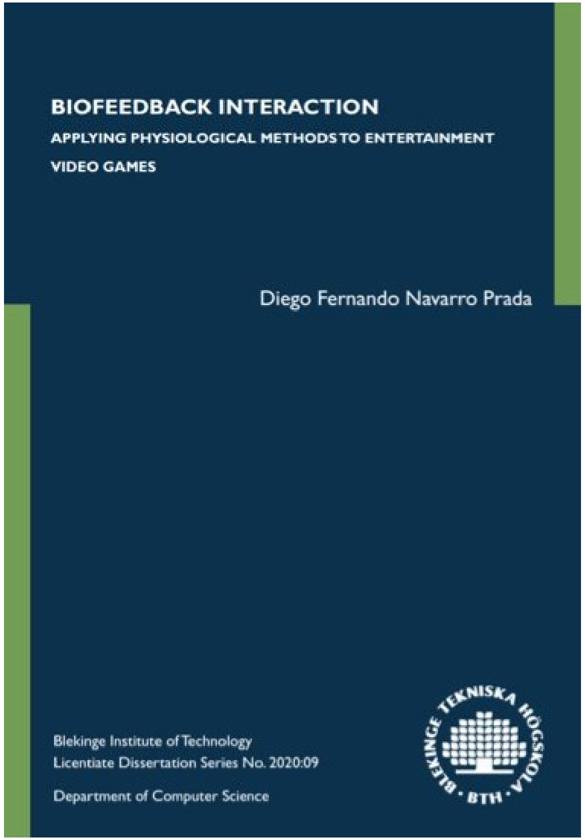
- I specialize in the design, development, and evaluation of **video games** and **real-time interactive technology**.

- Lecturer at **BTH** since **2014**:
  - Applied A.I., A.I. for games.
  - Large/small game project.
  - Plug-in construction and scripting languages.
  - Human-computer interaction (HCI).
  - Research orientation, Research methodology.
  - Construction of game editing tools.
  - Data science, Interactive data visualization.
  - 3D Programming III.
  - UNIX and GNU/Linux.
  - Bachelor/master thesis in digital game development.

Hi! I'm Diego

- Working on Scientific Research since 2009:
  - Psychophysiological feedback for video games and VR interaction.
  - HCI.
  - Interaction design.
  - A.I.
  - Interactive and scientific visualization.
  - Computer graphics.
  - Hardware and electronic architectures.
  - Quantum computing.

- Member of the HINTS research project.



Hi! I’m Diego

- Research:
  - Psychophysiological interaction for video games and VR.

Course Staff

Diego Fernando Navarro Prada.

- Course Responsible (structure, content, grades, Ladok).
- Management (schedules, canvas, discord, budgets, etc.).
- GNU/Linux + Shell Module Lecturer.
- Assignment 1 responsible (supervision, grading).

Course Staff

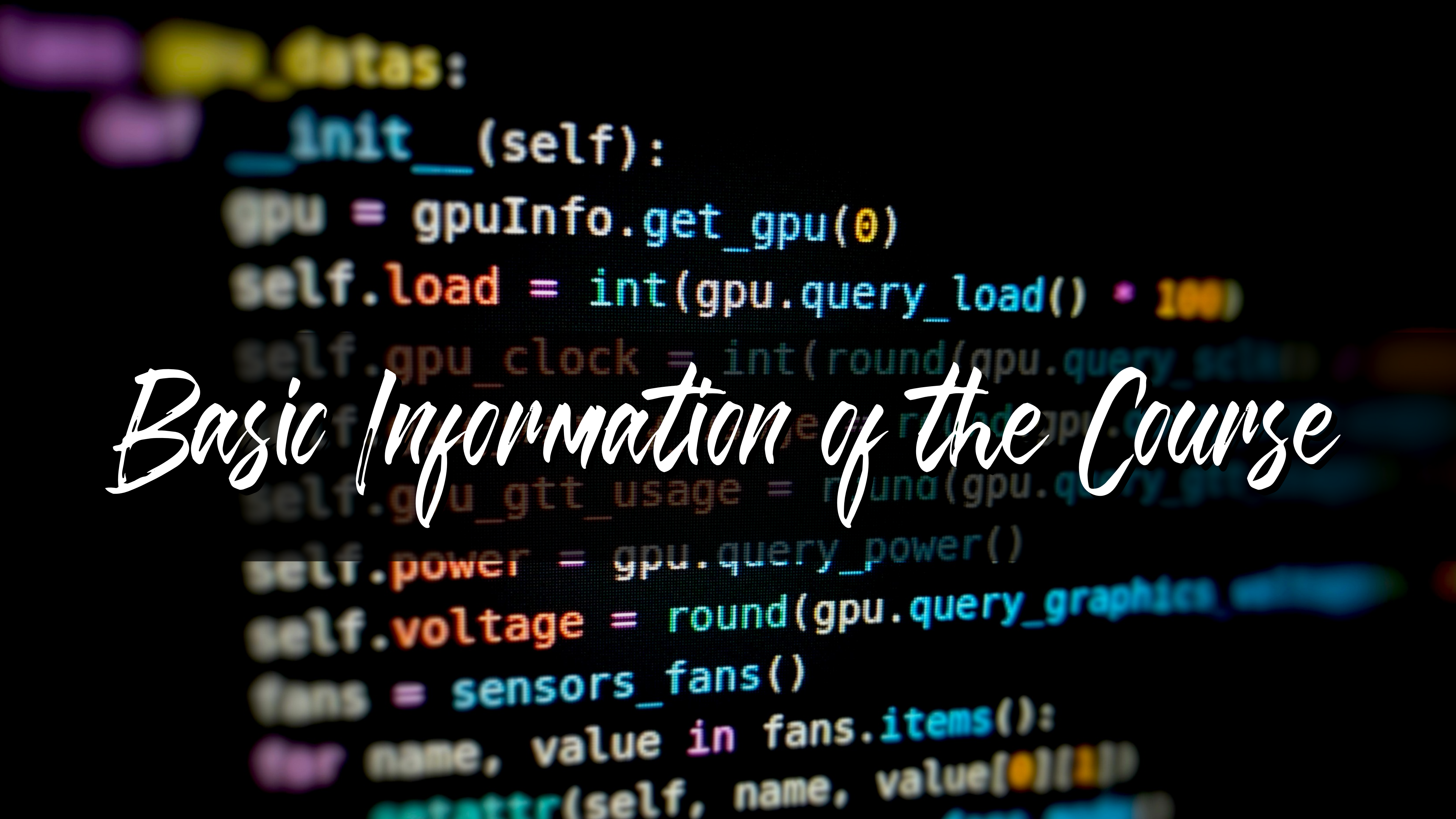
Amir Yavariabdi.

- C Module Lecturer.
- Assignment 2 responsible (supervision, grading).

Course Staff

Christoffer Åleskog

- Assembly Module Lecturer.
- Assignment 3 responsible (supervision, grading).



Basic Information of the Course

Basic information of the Course

- I was contacted by the planning department on September 2023, a month before going on a parental leave …

Basic information of the Course

Basic information of the Course

- How is the course currently doing?
  - Course evaluation 2022: 3.0/4.0
  - Course evaluation 2023: 3.3/4.0

- Student Feedback:
  - Course content was not connected with assignments.
  - Content and assignments were too advanced / basic.
  - Assignment grading and providing feedback was slow.
  - Material was insufficient.
  - Overall, better structure and support.

Basic information of the Course

- Course Prerequisites:
  - DV1457 (31 ECTS): Programming (15 ECTS), Algorithms and Data Structures (6 ECTS), Operative systems (6 ECTS), Data and Local Networks (4 ECTS).
  - DV1578 (28 ECTS): Programming (12 ECTS), Algorithms and Data Structures (6 ECTS), Operative Systems (6ECTS), Computer Communications and Local Networks (4 ECTS).

  - We will assume you have a fair experience in programming, that you are capable of designing, implementing, and debugging computational solutions, that you can use and understand technical documentation, and that you have experience accessing and monitoring processing threads and memory allocations.

Basic information of the Course

- Course Goals (Syllabus):
  1. Introduce the UNIX/Linux OS, and its components.
  2. Learn the basics of Shell programming (BASH).
  3. Use C as a system language for UNIX / Linux.
  4. Learn the basics of Assembly language.

- Our Main Goal:
  1. Understand the UNIX/Linux OS, and its programming interfaces (i.e. from high to low level).

Basic information of the Course

- Learning Activities:
  - Lectures (main theoretical component).
  - Labs (main practical component).
  - Course Assignments (course assessment):
    - 3 Assignments.

Basic information of the Course

- Credits
  - DV1457: 7.5 ECTS.
  - DV1578: 6 ECTS.
  - 1 ECTS ≃ 27 hours of presential and independent work.

- 162 - 202 hours approx (time needed to 100% Elden Ring twice).
  - 18 hours for 9 lectures.
  - 9 hours for 3 labs.
  - 24 hours for 4 supervision days.
  - 111-151 hours for assignment development.
    - Around 37-50 hours per assignment.

Basic information of the Course

- The course presents a **fair challenge** to students.
  - Exploration of the different programming interfaces to control and monitor the UNIX/Linux OS.

- **This course is of great importance**: This course presents a **realistic scenario** of the computational environment that are typical in computer security, web development, and general computer science.

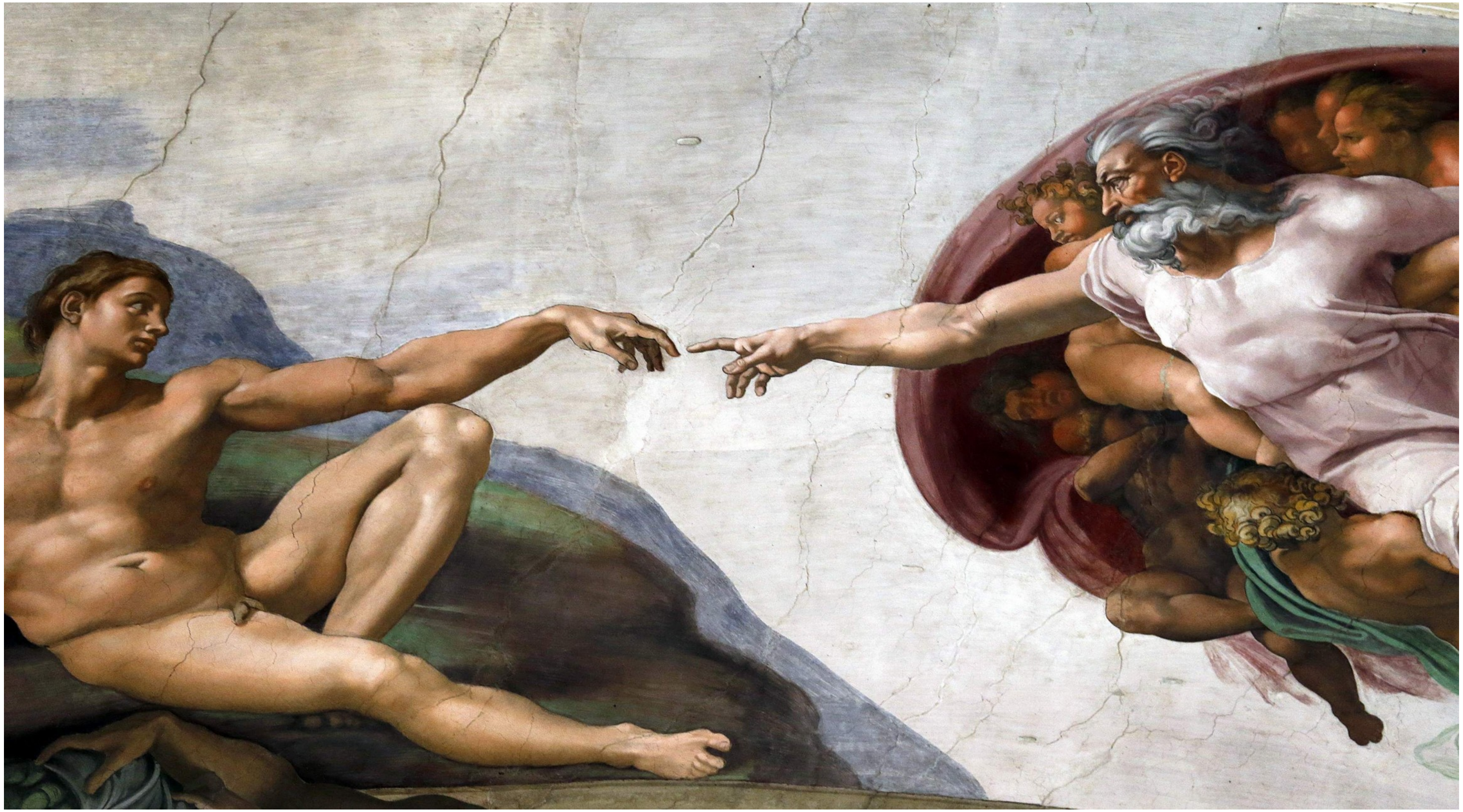
Basic information of the Course

- UNIX and Linux are the **most common computational environments** used in the areas of computer science, machine learning, computer security, and web (development and development).

- By mastering the use of UNIX and Linux, you will open yourself with a new set of computational tools and skills that are, arguably, **more robust, efficient, and versatile** than those offered by other computational environments.

Pedagogical Approach

Pedagogical Approach



Pedagogical Approach

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Pedagogical Approach

- Constructivist Approach.
  - Constructive Alignment.
  - Point-by-point lecturing.
  - Socratic Method.
  - Scaffolding.

- Lectures as the main teaching mechanism.
- Labs as the main empirical learning activities.
- Supervisions, Canvas, and Discord as reinforcement tools.

Pedagogical Approach

- Course Structure.
  - 9 Lectures.
  - 3 Labs.
  - 4 Supervisions.
  - 3 Assignments.

Week	Activity	Topic
36	Lecture	Introduction + UNIX/Linux Module
	Lecture	UNIX/Linux Module
	Lecture	UNIX/Linux Module
37	Lab	Shell Programming
	Supervision	
38	Lecture	C Module
	Lecture	C Module
	Assessment	Assignment 1
39	Lecture	C Module
	Lecture	C Module
40	Lab	C Programming in Unix/Linux
	Supervision	
41	Lecture	Assembly Module
	Lecture	Assembly Module
	Assessment	Assignment 2
42	Lab	Assembly Programming in Unix/Linux
	Supervision	
43	Supervision	
	Assessment	Assignment 3

Pedagogical Approach

- Lectures.
  - Approx. 100 minutes, with a 5-10 minute break.
  - Academic quarter, but try to be on time!
  - Attendance to Lectures is voluntary, except for Lecture 4 due to the 3-week roll-call.

Pedagogical Approach

- Labs.
  - Pre-recorded live coding sessions (due to number of student).
  - In-site support for the development of the lab.
  - Concept introduction (Lecture) → Implementation (Practical Module).
  - **Attendance and development** of the labs is **voluntary**.

Pedagogical Approach

- Supervisions.
  - Use Discord for general questions.
  - Book a supervision slot (15 min. approx.) for a more in-depth review, using the Excel sheet in Canvas.
  - Participation in supervisions is voluntary.

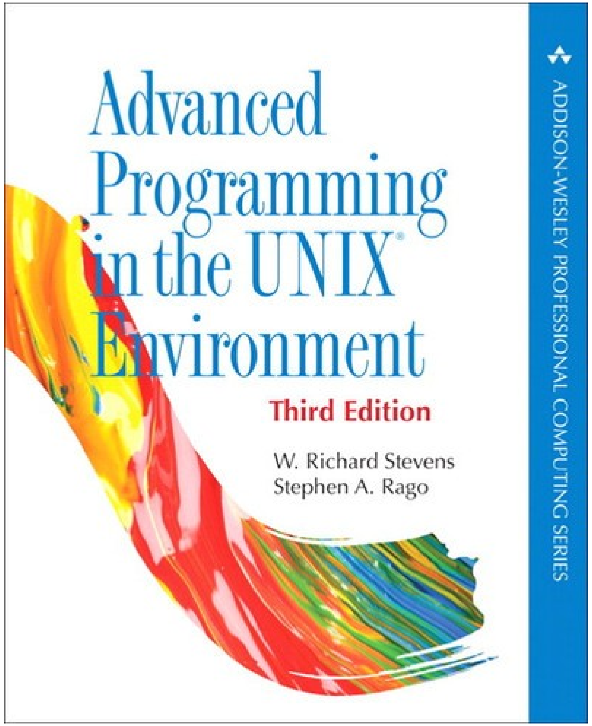
Pedagogical Approach

- Course Assessment.
  - 3 Assignments, developed in groups.
  - Up to 3 submission opportunities are offered for each assignment.
  - All assignment deadlines are posted in Canvas (submissions at 18:00).
  - Grades are uploaded to Ladok approximately a week after being published in Canvas.

Pedagogical Approach

- Bibliography.
  - "Advanced Programming in the UNIX Environment" (2013).

  - "The Linux Programming Interface" by Barrett. (2010).
  - "How Linux works" (2021).
  - "The Linux Command Line" (2019).



Closing Remarks

Closing Remarks

- This course is amazing! - *Diego Navarro, 2024*.
  - The course will provide you with the tools to take the most advantage out of the UNIX/Linux operative systems.
  - Give it your best shot!

- The course gives students a lot of freedom:
  - Participation in Lectures / Labs / Supervisions.
  - It is **up to you** to make the best out of the resources given to you!

Closing Remarks

- Mandatory moments:
  - 3 week roll-call (Week 38: September 16th).
- All **deadlines are strong deadlines!**
  - Late submissions will not be accepted.
  - Submissions are only accepted via Canvas.
- **Zero tolerance for <u>Plagiarism</u>!**
  - Submissions are automatically checked by Ouriginal.

Closing Remarks

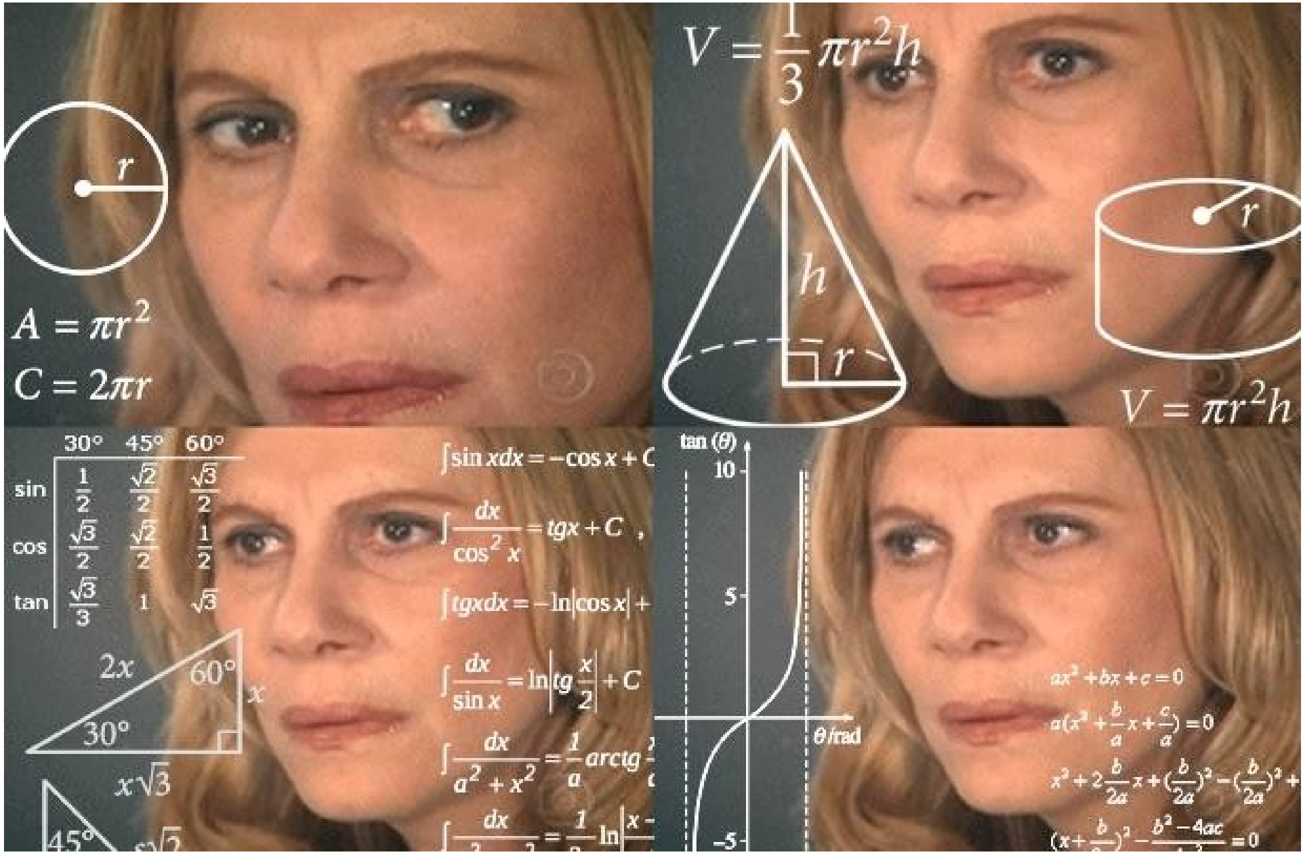
- **LLMs** (e.g. ChatGPT) are useful tools that, unfortunately, could prevent you from develop comprehensively your own potential.

- It is not forbidden to use LLMs in the course (you guys are going to use them despite). However use them appropriately:
  - Don't use an LLM to develop the assignment for you, but to help you **understand concepts/code** that may be unclear.

Closing Remarks

- We have a Discord server!
  - Ease communication and collaboration between all course members!
- Canvas is the official means of communication for the course.
  - All course information and learning material will be in Canvas.
- Special needs:
  - Contact funka@bth.se
  - Talk to us if you need any help!

Questions?



$A = \pi r^2$
$C = 2\pi r$
$V = \frac{1}{3}\pi r^2 h$
$V = \pi r^2 h$
r
h
30° 45° 60°
sin
cos
tan
2x
60°
30°
$x\sqrt{3}$
45°
tan (θ)
θ/rad
10
5
−5
$ax^2 + bx + c = 0$